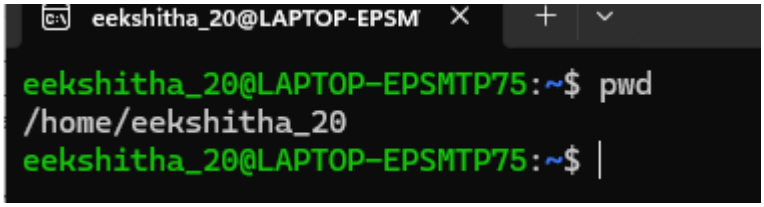
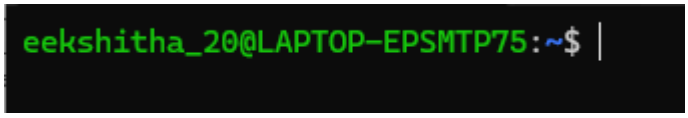
*Topic 1: Basic Linux Commands

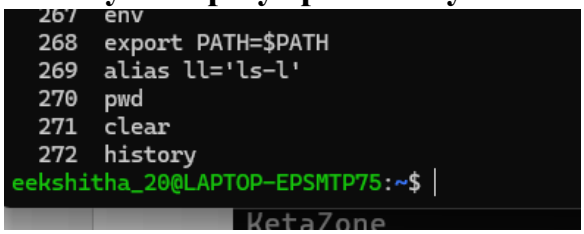
1. **Pwd** ~ Display the current working directory



2. **Clear ~ clear the terminal screen**



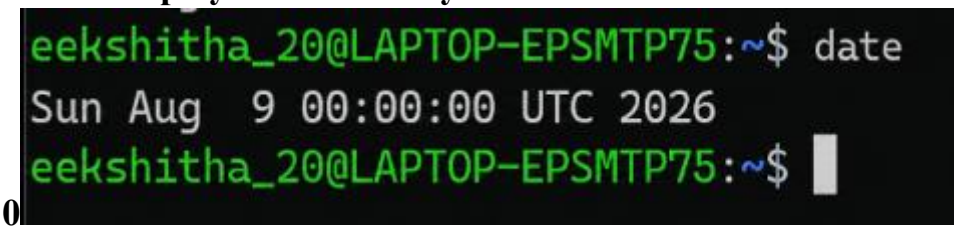
3. **History ~ displays previously executed commands**



4. **Whoami ~ display the current logged-in username**

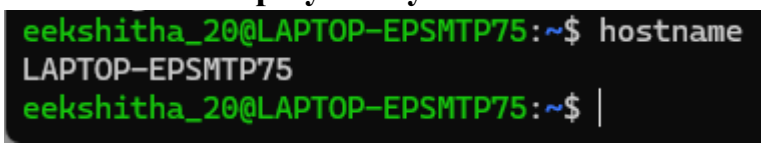


5. **Date ~ display the current system date and time**

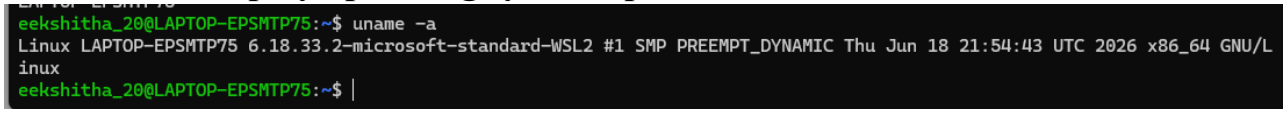


0

6. **Hostname ~ Display the system hostname**



7. **Uname -a ~ Display operating system operation**



8. **Exit ~ exit the terminal**



TOPIC 2 ~ DIRECTORY MANAGEMENT

1. **Ls ~ list files and directories**

```
eekshitha_20@LAPTOP-EPSMTP75:~$ ls
CPrograms       FifthProgram     FourthProgram     SeventhProgram     SystemCall          ThrdProgram.c        frstProgram        secProgram
DisplayPID      FifthProgram.c   FourthProgram.c   SeventhProgram.c   SystemCall.c        eekshitha            frstProgram.c      secProgram.c
DisplayPID.c    FirstProgram.c   NinthProgram      Shellcommand.c     SystemCallParent    exaample.txt         myPrograms         shellcommand
EightProgram    ForkProgram      NinthProgram.c    SixthProgram       SystemCallParent.c  firProgram.c         s2Commands         shellcommand.c
EightProgram.c  ForkProgram.c    NonthProgram      SixthProgram.c     ThrdProgram         firstProgram.c.save  s2SampleProgram.c
```

2. **Ls -a ~ display detailed file information**

```
eekshitha_20@LAPTOP-EPSMTP75:~$ ls -a
.               .local          EightProgram.c  FourthProgram.c  SixthProgram       ThrdProgram.c        myPrograms
..              .motd_shown     FifthProgram    NinthProgram     SixthProgram.c     eekshitha            s2Commands
.bash_history   .profile        FifthProgram.c  NinthProgram.c   SystemCall         exaample.txt         s2SampleProgram.c
.bash_logout    CPrograms       FirstProgram.c  NonthProgram     SystemCall.c       firProgram.c         secProgram
.bashrc         DisplayPID      ForkProgram     SeventhProgram   SystemCallParent   firstProgram.c.save  secProgram.c
.cache          DisplayPID.c    ForkProgram.c   SeventhProgram.c SystemCallParent.c frstProgram          shellcommand
.config         EightProgram    FourthProgram   Shellcommand.c   ThrdProgram        frstProgram.c        shellcommand.c
```

3. **Cd ~ change the current directory**

```
eekshitha_20@LAPTOP-EPSMTP75:~$ cd foldername
-bash: cd: foldername: No such file or directory
eekshitha_20@LAPTOP-EPSMTP75:~$ cd FirstProgram.c
-bash: cd: FirstProgram.c: Not a directory
```

4. **Cd ..~ move to parent directory**

```
eekshitha_20@LAPTOP-EPSMTP75:~$ cd..
cd..: command not found
```

5. **mkdir ~ create a new directory**

```
eekshitha_20@LAPTOP-EPSM
eekshitha_20@LAPTOP-EPSMTP75:~$ mkdir OSSP
eekshitha_20@LAPTOP-EPSMTP75:~$
```

6. **rmdir ~ remove a directory**

```
eekshitha_20@LAPTOP-EPSMTP75:~$ rmdir OSSP
eekshitha_20@LAPTOP-EPSMTP75:~$
```

7. **find FileName.c ~ search for files and directories**

```
eekshitha_20@LAPTOP-EPSMTP75:~$ find FirstProgram.c
FirstProgram.c
```

8. locate ~ Quickly locate files

```
eekshitha_20@LAPTOP-EPSMTP75:~$ locate FirstProgram.c
/var/lib/plocate/plocate.db: No such file or directory
```

9.ls -l

```
eekshitha_20@LAPTOP-EPSMTP75:~$ ls -l
total 332
drwxr-xr-x 2 eekshitha_20 eekshitha_20  4096 Aug  4 17:30 CPrograms
-rwxr-xr-x 1 eekshitha_20 eekshitha_20 16128 Aug  4 15:51 DisplayPID
-rw-r--r-- 1 eekshitha_20 eekshitha_20   499 Aug  4 15:50 DisplayPID.c
-rwxr-xr-x 1 eekshitha_20 eekshitha_20 16000 Aug  3 15:14 EightProgram
-rw-r--r-- 1 eekshitha_20 eekshitha_20   160 Aug  3 15:02 EightProgram.c
-rwxr-xr-x 1 eekshitha_20 eekshitha_20 16000 Aug  2 16:55 FifthProgram
-rw-r--r-- 1 eekshitha_20 eekshitha_20   231 Aug  2 16:54 FifthProgram.c
-rw-r--r-- 1 eekshitha_20 eekshitha_20    78 Aug  4 13:16 FirstProgram.c
-rwxr-xr-x 1 eekshitha_20 eekshitha_20 16000 Aug  4 14:56 ForkProgram
-rw-r--r-- 1 eekshitha_20 eekshitha_20   310 Aug  4 14:56 ForkProgram.c
-rwxr-xr-x 1 eekshitha_20 eekshitha_20 16000 Jul 31 10:02 FourthProgram
-rw-r--r-- 1 eekshitha_20 eekshitha_20   104 Jul 31 10:01 FourthProgram.c
-rwxr-xr-x 1 eekshitha_20 eekshitha_20 16088 Aug  3 15:20 NinthProgram
-rw-r--r-- 1 eekshitha_20 eekshitha_20   264 Aug  3 15:17 NinthProgram.c
-rwxr-xr-x 1 eekshitha_20 eekshitha_20 16088 Aug  3 15:19 NonthProgram
-rwxr-xr-x 1 eekshitha_20 eekshitha_20 16040 Aug  2 17:06 SeventhProgram
-rw-r--r-- 1 eekshitha_20 eekshitha_20   275 Aug  2 17:05 SeventhProgram.c
-rw-r--r-- 1 eekshitha_20 eekshitha_20   182 Aug  4 15:20 Shellcommand.c
-rwxr-xr-x 1 eekshitha_20 eekshitha_20 16088 Aug  2 17:02 SixthProgram
-rw-r--r-- 1 eekshitha_20 eekshitha_20   288 Aug  2 16:58 SixthProgram.c
-rwxr-xr-x 1 eekshitha_20 eekshitha_20 16184 Aug  4 15:34 SystemCall
-rw-r--r-- 1 eekshitha_20 eekshitha_20   268 Aug  4 15:34 SystemCall.c
-rwxr-xr-x 1 eekshitha_20 eekshitha_20 16048 Aug  4 15:47 SystemCallParent
-rw-r--r-- 1 eekshitha_20 eekshitha_20   407 Aug  4 15:46 SystemCallParent.c
-rwxr-xr-x 1 eekshitha_20 eekshitha_20 16000 Jul 31 09:59 ThrdProgram
-rw-r--r-- 1 eekshitha_20 eekshitha_20   111 Jul 31 09:58 ThrdProgram.c
drwxr-xr-x 2 eekshitha_20 eekshitha_20  4096 Jul 28 04:09 eekshitha
-rw-r--r-- 1 eekshitha_20 eekshitha_20     0 Jul 31 04:01 exaample.txt
-rw-r--r-- 1 eekshitha_20 eekshitha_20    96 Jul 31 09:30 firProgram.c
-rw------- 1 eekshitha_20 eekshitha_20    94 Jul 31 09:35 firstProgram.c.save
-rwxr-xr-x 1 eekshitha_20 eekshitha_20 15960 Jul 31 09:38 frstProgram
-rw-r--r-- 1 eekshitha_20 eekshitha_20    80 Jul 31 09:38 frstProgram.c
drwxr-xr-x 2 eekshitha_20 eekshitha_20  4096 Aug  4 17:09 myPrograms
drwxr-xr-x 2 eekshitha_20 eekshitha_20  4096 Jul 31 04:03 s2Commands
-rw-r--r-- 1 eekshitha_20 eekshitha_20    89 Jul 31 09:15 s2SampleProgram.c
-rwxr-xr-x 1 eekshitha_20 eekshitha_20 16064 Jul 31 09:45 secProgram
-rw-r--r-- 1 eekshitha_20 eekshitha_20   159 Jul 31 09:45 secProgram.c
-rwxr-xr-x 1 eekshitha_20 eekshitha_20 16064 Aug  4 15:22 shellcommand
-rw-r--r-- 1 eekshitha_20 eekshitha_20   182 Aug  4 15:21 shellcommand.c
```

3.FILE MANAGEMENT

1.touch ~ Create new empty file

```
eekshitha_20@LAPTOP-EPSMTP75:~$ touch KL.c
eekshitha_20@LAPTOP-EPSMTP75:~$
```

2. cat ~ display file contents

```
eekshitha_20@LAPTOP-EPSMTP75:~$ cat FirstProgram.c
#include<stdio.h>

int main()
{
    printf("Hello Linux\n");
    return 0;
}
```

3. cp ~ copy files or directories

```
eekshitha_20@LAPTOP-EPSMTP75:~$ cp FirstProgram.c FirstFile.c
eekshitha_20@LAPTOP-EPSMTP75:~$
```

4. mv ~ move or rename files

```
eekshitha_20@LAPTOP-EPSMTP75:~$ mv FirstProgram.c helllo.c
eekshitha_20@LAPTOP-EPSMTP75:~$
```

5. rm ~ remove file

```
ketana_sindhu@KetaZone:~$ rm hello.c
ketana_sindhu@KetaZone:~$
```

6. head ~ display first lines of a file

```
eekshitha_20@LAPTOP-EPSMTP75:~$ head FirstFile.c
#include<stdio.h>

int main()
{
    printf("Hello Linux\n");
    return 0;
}
```

7. tail ~ display last lines of the file

```
eekshitha_20@LAPTOP-EPSMTP75:~$ tail FirstFile.c
#include<stdio.h>

int main()
{
    printf("Hello Linux\n");
    return 0;
}
```

8.more ~ view a file page by page

```
eekshitha_20@LAPTOP-EPSMTP75:~$ more FirstFile.c
#include<stdio.h>

int main()
{
    printf("Hello Linux\n");
    return 0;
}
```

9.less ~ Scroll through a file

```
#include<stdio.h>

int main()
{
    printf("Hello Linux\n");
    return 0;
}
FirstFile.c (END)
```

10.wc ~ count lines, words and characters

```
eekshitha_20@LAPTOP-EPSMTP75:~$ wc FirstFile.c
 8  9 78 FirstFile.c
eekshitha_20@LAPTOP-EPSMTP75:~$
```

11.diff ~ compare two files

```
1a2
> #include<unistd.h>
5c6,19
<     printf("Hello Linux\n");
---
>     pid_t pid;
>
>     pid=fork();
>
>     if(pid==0)
>     {
>         printf("Child PID = %d\n",getpid());
>         printf("Parent PID = %d\n",getppid());
>     }
>     else
>     {
>         printf("Parent PID = %d\n",getpid());
>     }
>
```

12.sort ~ sort file contents

```
eekshitha_20@LAPTOP-EPSMTP75:~$ sort FirstFile.c


    printf("Hello Linux\n");
    return 0;
#include<stdio.h>
int main()
{
}
```

13.uniq ~ remove duplicate files

```
eekshitha_20@LAPTOP-EPSMTP75:~$ uniq FirstFile.c
#include<stdio.h>

int main()
{
    printf("Hello Linux\n");
    return 0;
}
```

14.grep ~ search for text inside a file

```
eekshitha_20@LAPTOP-EPSMTP75:~$ grep main FirstFile.c
int main()
eekshitha_20@LAPTOP-EPSMTP75:~$
```

4. TOPIC ~ FILE EDITING

1.nano ~ create or edit a file

```
eekshitha_20@LAPTOP-EPSMTP75:~$ nano Hello.c
eekshitha_20@LAPTOP-EPSMTP75:~$
```

2. vi ~ edit a file using vim editor

```
#include<stdio.h>

int main()
{
    printf("Hello Linux\n");
    return 0;
}
```

5. C PROGRAM COMPILATION AND EXECUTION

1. **gcc ~ compile a C program**

```
eekshitha_20@LAPTOP-EPSMTP75:~$ gcc FirstFile.c -o file
eekshitha_20@LAPTOP-EPSMTP75:~$ ./file
```

2. **. /program ~ Execute the compiled program**

```
eekshitha_20@LAPTOP-EPSM
eekshitha_20@LAPTOP-EPSMTP75:~$ nano FirstFile.c
eekshitha_20@LAPTOP-EPSMTP75:~$ gcc FirstFile.c -o file
eekshitha_20@LAPTOP-EPSMTP75:~$ ./file
Hello Linux
```

3. **which ~ Locate the executable of a command**

```
eekshitha_20@LAPTOP-EPSMTP75:~$ which gcc
/usr/bin/gcc
eekshitha_20@LAPTOP-EPSMTP75:~$
```

4. **whereis ~ Locate binary, source and manual pages**

```
eekshitha_20@LAPTOP-EPSMTP75:~$ whereis gcc
gcc: /usr/bin/gcc /usr/lib/gcc /usr/libexec/gcc /usr/share/gcc /usr/share/man/man1/gcc
eekshitha_20@LAPTOP-EPSMTP75:~$
```

5. **man ~ display manual pages**

```
eekshitha_20@LAPTOP-EPSM
GCC(1)                                  GNU                                  GCC(1)

NAME
       gcc - GNU project C and C++ compiler

SYNOPSIS
       gcc [-c|-S|-E] [-std=standard]
           [-g] [-pg] [-Olevel]
           [-Wwarn...] [-Wpedantic]
           [-Idir...] [-Ldir...]
           [-Dmacro[=defn]...] [-Umacro]
           [-foption...] [-mmachine-option...]
           [-o outfile] [@file] infile...

       Only the most useful options are listed here; see below for the remainder.  g++ accepts mostly the same options as gcc.

DESCRIPTION
       When you invoke GCC, it normally does preprocessing, compilation, assembly and linking.  The "overall options" allow you to stop this
       process at an intermediate stage.  For example, the -c option says not to run the linker.  Then the output consists of object files output
       by the assembler.

       Other options are passed on to one or more stages of processing.  Some options control the preprocessor and others the compiler itself.  Yet
       other options control the assembler and linker; most of these are not documented here, since you rarely need to use any of them.

       Most of the command-line options that you can use with GCC are useful for C programs; when an option is only useful with another language
       (usually C++), the explanation says so explicitly.  If the description for a particular option does not mention a source language, you can
       use that option with all supported languages.

       The usual way to run GCC is to run the executable called gcc, or machine-gcc when cross-compiling, or machine-gcc-version to run a specific
       version of GCC.  When you compile C++ programs, you should invoke GCC as g++ instead.

       The gcc program accepts options and file names as operands.  Many options have multi-letter names; therefore multiple single-letter options
       may not be grouped: -dv is very different from -d -v.

Manual page gcc(1) line 1 (press h for help or q to quit)
```

6.PROCESS MANAGEMENT

1. ps ~ display running processes

```
eekshitha_20@LAPTOP-EPSMTP75:~$ ps
    PID TTY          TIME CMD
    831 pts/0    00:00:00 bash
    859 pts/0    00:00:00 ps
eekshitha_20@LAPTOP-EPSMTP75:~$
```

2. ps -ef ~ display all running processes

```
eekshitha_20@LAPTOP-EPSMTP75:~$ ps -ef
UID          PID    PPID  C STIME TTY          TIME CMD
root           1       0  0 09:26 ?        00:00:01 /sbin/init
root           2       1  0 09:26 hvc0     00:00:00 /init
root           6       2  0 09:26 hvc0     00:00:00 plan9 --control-socket 7 --log-level 4 --server-fd 8 --pipe-fd 10 --
root          49       1  0 09:26 ?        00:00:00 /usr/lib/systemd/systemd-journald
systemd+      81       1  0 09:26 ?        00:00:00 /usr/lib/systemd/systemd-resolved
root          91       1  0 09:26 ?        00:00:01 /usr/lib/systemd/systemd-udevd
root         146       1  0 09:26 ?        00:00:00 /bin/sh /usr/lib/systemd/scripts/chronyd-starter.sh -n -F 1
root         148       1  0 09:26 ?        00:00:00 /usr/sbin/cron -f -P
message+     151       1  0 09:26 ?        00:00:00 @dbus-daemon --system --address=systemd: --nofork --nopidfile --syst
root         160       1  0 09:26 ?        00:00:00 /usr/bin/python3 /usr/bin/networkd-dispatcher --run-startup-triggers
root         171       1  0 09:26 ?        00:00:00 /usr/lib/systemd/systemd-logind
root         191       1  0 09:26 tty1     00:00:00 /usr/sbin/agetty --noreset --noclear --issue-file=/etc/issue:/etc/is
syslog       194       1  0 09:26 ?        00:00:00 /usr/sbin/rsyslogd -n -iNONE
root         220       1  0 09:26 ?        00:00:00 /usr/bin/python3 /usr/share/unattended-upgrades/unattended-upgrade-s
_chrony      235     146  0 09:26 ?        00:00:00 /usr/sbin/chronyd -n -F 1 -x
_chrony      266     235  0 09:26 ?        00:00:00 /usr/sbin/chronyd -n -F 1 -x
root         383       2  0 09:26 ?        00:00:00 login -- eekshitha_20
eekshit+     431       1  0 09:26 ?        00:00:00 /usr/lib/systemd/systemd --user
eekshit+     433     431  0 09:26 ?        00:00:00 (sd-pam)
eekshit+     459     383  0 09:26 pts/1    00:00:00 -bash
root         823       2  0 09:38 ?        00:00:00 /init
root         824     823  0 09:38 ?        00:00:00 /init
eekshit+     831     824  0 09:38 pts/0    00:00:00 -bash
eekshit+     885     831  0 09:40 pts/0    00:00:00 ps -ef
eekshitha_20@LAPTOP-EPSMTP75:~$
```

3. top ~ monitor running processes

```
eekshitha_20@LAPTOP-EPSM
top - 09:47:55 up 18 min,  1 user,  load average: 0.03, 0.03, 0.00
Tasks:  24 total,   1 running,  23 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni,100.0 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   3736.6 total,   3142.1 free,    468.0 used,    202.0 buff/cache
MiB Swap:   1024.0 total,   1024.0 free,      0.0 used.   3268.6 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0   24460  15504  11580 S   0.0   0.4   0:01.08 systemd
      2 root      20   0    3180   2204   2072 S   0.0   0.1   0:00.21 init-systemd(Ub
      6 root      20   0    3180   2048   1940 S   0.0   0.1   0:00.00 init
     49 root      19  -1   50400  16828  15540 S   0.0   0.4   0:00.50 systemd-journal
     81 systemd+  20   0   22540  14560  11940 S   0.0   0.4   0:00.15 systemd-resolve
     91 root      20   0   35296  12276   9256 S   0.0   0.3   0:01.67 systemd-udevd
    146 root      20   0    2888   1944   1820 S   0.0   0.1   0:00.06 chronyd-starter
    148 root      20   0    4472   3172   2924 S   0.0   0.1   0:00.04 cron
    151 message+  20   0    8820   5380   4716 S   0.0   0.1   0:00.10 dbus-daemon
    160 root      20   0   44092  29808  14664 S   0.0   0.8   0:00.29 networkd-dispat
    171 root      20   0   18432   9436   8192 S   0.0   0.2   0:00.12 systemd-logind
    191 root      20   0    5492   2712   2528 S   0.0   0.1   0:00.01 agetty
    194 syslog    20   0  220548   5884   4576 S   0.0   0.2   0:00.14 rsyslogd
    220 root      20   0  123456  32708  18096 S   0.0   0.9   0:00.25 unattended-upgr
    235 _chrony   20   0   24204  11708   7140 S   0.0   0.3   0:00.13 chronyd
    266 _chrony   20   0   12092   2404   1760 S   0.0   0.1   0:00.01 chronyd
    383 root      20   0    8648   5504   4660 S   0.0   0.1   0:00.00 login
    431 eekshit+  20   0   22332  13560  10956 S   0.0   0.4   0:00.16 systemd
    433 eekshit+  20   0   22912   4124   2120 S   0.0   0.1   0:00.00 (sd-pam)
    459 eekshit+  20   0    6136   5480   3928 S   0.0   0.1   0:00.03 bash
    823 root      20   0    3184   1116    980 S   0.0   0.0   0:00.00 SessionLeader
    824 root      20   0    3200   1256   1108 S   0.0   0.0   0:00.00 Relay(831)
    831 eekshit+  20   0    6268   5548   3872 S   0.0   0.1   0:00.09 bash
    940 eekshit+  20   0    8160   5988   3896 R   0.0   0.2   0:00.00 top
```

4. kill ~ terminate a process

```
eekshitha_20@LAPTOP-EPSMTP75:~$ kill 1234
-bash: kill: (1234) - No such process
eekshitha_20@LAPTOP-EPSMTP75:~$
```

5.kill -9 ~ Forcefully terminate a process

```
eekshitha_20@LAPTOP-EPSMTP75:~$ kill -9
kill: usage: kill [-s sigspec | -n signum | -sigspec] pid | jobspec ... or kill -l [sigspec]
eekshitha_20@LAPTOP-EPSMTP75:~$
```

7.JOB CONTROL

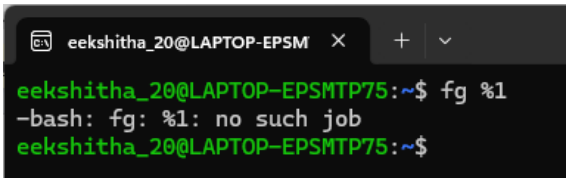
1. jobs ~ Display background jobs

```
eekshitha_20@LAPTOP-EPSMTP75:~$ jobs
eekshitha_20@LAPTOP-EPSMTP75:~$
```

2. bg ~ Resume a job in the background

```
eekshitha_20@LAPTOP-EPSMTP75:~$ bg %1
-bash: bg: %1: no such job
eekshitha_20@LAPTOP-EPSMTP75:~$
```

3.fg ~ Bring a job to the foreground



4.& - Run a process in the background

```
eekshitha_20@LAPTOP-EPSMTP75:~$ ./FirstFile &
[1] 521
-bash: ./FirstFile: No such file or directory
[1]+  Exit 127                 ./FirstFile
eekshitha_20@LAPTOP-EPSMTP75:~$
```

8.FILE PERMISSIONS AND LINKS

1. chmod ~ Change file permissions

```
eekshitha_20@LAPTOP-EPSMTP75:~$ chomde +x script.sh
chomde: command not found
eekshitha_20@LAPTOP-EPSMTP75:~$
```

2. chmod 755 ~ Set rad, write and execute permissions

```
chomde: command not found
eekshitha_20@LAPTOP-EPSMTP75:~$ chomd 755
Command 'chomd' not found, did you mean:
```

3.ln ~ create a hard link

```
eekshitha_20@LAPTOP-EPSMTP75:~$ ln FirstFile.c HELLO.c
```

4.ln -s ~ create a symbolic link

```
eekshitha_20@LAPTOP-EPSMTP75:~$ ln -s FirstFile shortcut
eekshitha_20@LAPTOP-EPSMTP75:~$
```

9. DISK AND MEMORY INFORMATION

1. df -h ~ display disk space usage

```
Filesystem      Size  Used Avail Use% Mounted on
none            1.9G     0  1.9G   0% /usr/lib/modules/6.18.33.2-microsoft-standard-WSL2
none            1.9G  4.0K  1.9G   1% /mnt/wsl
drivers         476G  152G  325G  32% /usr/lib/wsl/drivers
/dev/sdd       1007G  2.1G  954G   1% /
none            1.9G   36K  1.9G   1% /mnt/wslg
none            1.9G     0  1.9G   0% /usr/lib/wsl/lib
rootfs          1.9G  2.8M  1.9G   1% /init
none            1.9G  532K  1.9G   1% /run
none            1.9G     0  1.9G   0% /run/lock
none            1.9G     0  1.9G   0% /run/shm
none            1.9G   80K  1.9G   1% /mnt/wslg/versions.txt
none            1.9G   80K  1.9G   1% /mnt/wslg/doc
C:\             476G  152G  325G  32% /mnt/c
none            1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
tmpfs           1.9G     0  1.9G   0% /tmp
none            1.0M     0  1.0M   0% /run/credentials/systemd-resolved.service
none            1.0M     0  1.0M   0% /run/credentials/getty@tty1.service
tmpfs           374M   12K  374M   1% /run/user/1000
```

2. du -sh ~ display directory size

```
eekshitha_20@LAPTOP-EPSMTP75:~$ du -sh
496K    .
eekshitha_20@LAPTOP-EPSMTP75:~$
```

3. free -h ~ display memory usage (Linux)

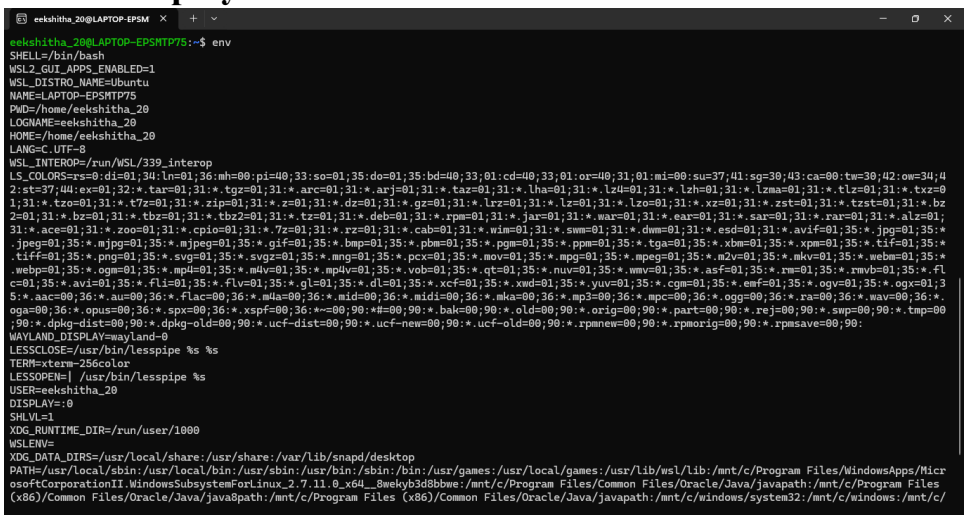
```
eekshitha_20@LAPTOP-EPSMTP75:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           3.6Gi       460Mi       3.1Gi       3.5Mi       170Mi       3.2Gi
Swap:          1.0Gi          0B       1.0Gi
eekshitha_20@LAPTOP-EPSMTP75:~$
```

10.USER AND ENVIRONMENT MANAGEMENT

1.id ~ display user and group IDs

```
eekshitha_20@LAPTOP-EPSMTP75:~$ id
uid=1000(eekshitha_20) gid=1000(eekshitha_20) groups=1000(eekshitha_20),4(adm),24(cdrom),27(sudo),30(dip),46(plugdev),100(users)
eekshitha_20@LAPTOP-EPSMTP75:~$
```

2.env ~ display environment variables



3.export ~ Set an environment variable

```
eekshitha_20@LAPTOP-EPSMTP75:~$ export PATH=$PATH
eekshitha_20@LAPTOP-EPSMTP75:~$
```

4. alias ~ Create a command shortcut

```
eekshitha_20@LAPTOP-EPSMTP75:~$ alias ll='ls-l'
eekshitha_20@LAPTOP-EPSMTP75:~$
```